

Cluster-forming $t > 3.24$ ($p < .001$) · cluster-extent FWE $p < .05$

model7 · $n = 58$ · surviving extent, % of mask: RPE +50.1 / -0.1 uRPE +2.0 / -16.7 Surprise +1.0 / -0.0

model7 — one GLM carrying all three signals: choice (surprise, value), feedback (uRPE first, then signed RPE). Because SPM orthogonalises serially within an event, uRPE keeps all variance it shares with signed RPE, and the RPE map is the residual. Apples-to-apples across signals.

a

			Covered by map						Effect in ROI			Are they different?		
			RPE		uRPE		Surprise		RPE	uRPE	Surprise	uRPE vs Surprise	RPE vs uRPE	RPE vs Surprise
			+	-	+	-	+	-						
RPE+ & uRPE- & Surprise+ * — 5 clusters														
1	DLPFC L	found by uRPE -	■	•	•	□	■	•	▲	▼	▲	▼	▲	▲
2	Middle temporal gyrus L	found by Surprise +	■	•	•	□	■	•	▲	▼	▲	▼	▲	▲
3	Precuneus L	found by Surprise +	■	•	•	□	■	•	▲	▼	▲	▼	▲	▲
4	DLPFC L (2)	found by Surprise +	■	•	•	□	■	•	▲	▼	▲	▼	▲	▲
5	Angular gyrus R	found by Surprise +	■	•	•	□	■	•	▲	▼	▲	▼	▲	▲
RPE+ & uRPE- * — 4 clusters														
6	Caudate / putamen R	found by RPE +	■	•	•	□	■	•	▲	▼	▲	▼	▲	▲
7	Caudate / putamen R (2)	found by uRPE -	■	•	•	□	■	•	▲	▼	▲	▼	▲	▲
8	Middle temporal gyrus R	found by uRPE -	■	•	•	□	■	•	▲	▼	▲	▼	▲	▲
9	Inferior parietal lobule L	found by uRPE -	■	•	•	□	■	•	▲	▼	▲	▼	▲	—
RPE- & uRPE+ * — 1 cluster														
10	Insula R (2)	found by RPE -	•	□	■	•	•	•	▼	▲	▲	▲	▼	▼
only uRPE+ — 3 clusters														
11	Insula R	found by uRPE +	•	■	■	•	•	•	▼	▲	▲	▲	▼	○
12	dmPFC R	found by uRPE +	■	•	■	•	•	•	▼	▲	▲	▲	▼	○
13	Insula L	found by uRPE +	•	•	■	•	•	•	▼	▲	▲	▲	—	○
RPE+ & uRPE+ — 1 cluster														
14	Inferior parietal lobule R	found by uRPE +	■	•	■	•	•	•	▲	▲	▼	▲	○	▲

Profile groups — 5 of them across 14 ROIs

RPE+ & uRPE- & Surprise+ * (5) RPE+ & uRPE- * (4) RPE- & uRPE+ * (1) only uRPE+ (3) RPE+ & uRPE+ (1)

* the same tissue responds positively to one signal and negatively to another — these groups are listed first.
Membership is map coverage: a map counts if it covers at least a tenth of the ROI. It is not the difference tests — a signal can be in the tag and still be “undetermined” against another.

Rows	Covered by map	Effect in ROI	Are they different?
One cluster found by one of the six contrasts. Rows are boxed by profile — the set of maps covering at least a tenth of them — and the box is drawn in that profile's colour. The number keys the row to its peak on the map pages.	Share of the ROI's voxels inside that map's surviving clusters.	Group one-sample t of that signal's contrast value, averaged over the ROI.	One signal against another inside the ROI, paired across subjects.
	100% of the ROI 50% of the ROI 10% of the ROI Open square = negative tail Dot = below 1%, no overlap	▲ Positive, and significant ▼ Negative, and significant △ Open = not significant Bigger marker = a larger effect. Holm-corrected over all ROI × signal tests.	▲ Different; points to the larger ○ Equivalent ($BF_{01} > 3$) — Undetermined TOST bound in the table is $d_z = 0.368$, the smallest effect this design can find with 80% power.

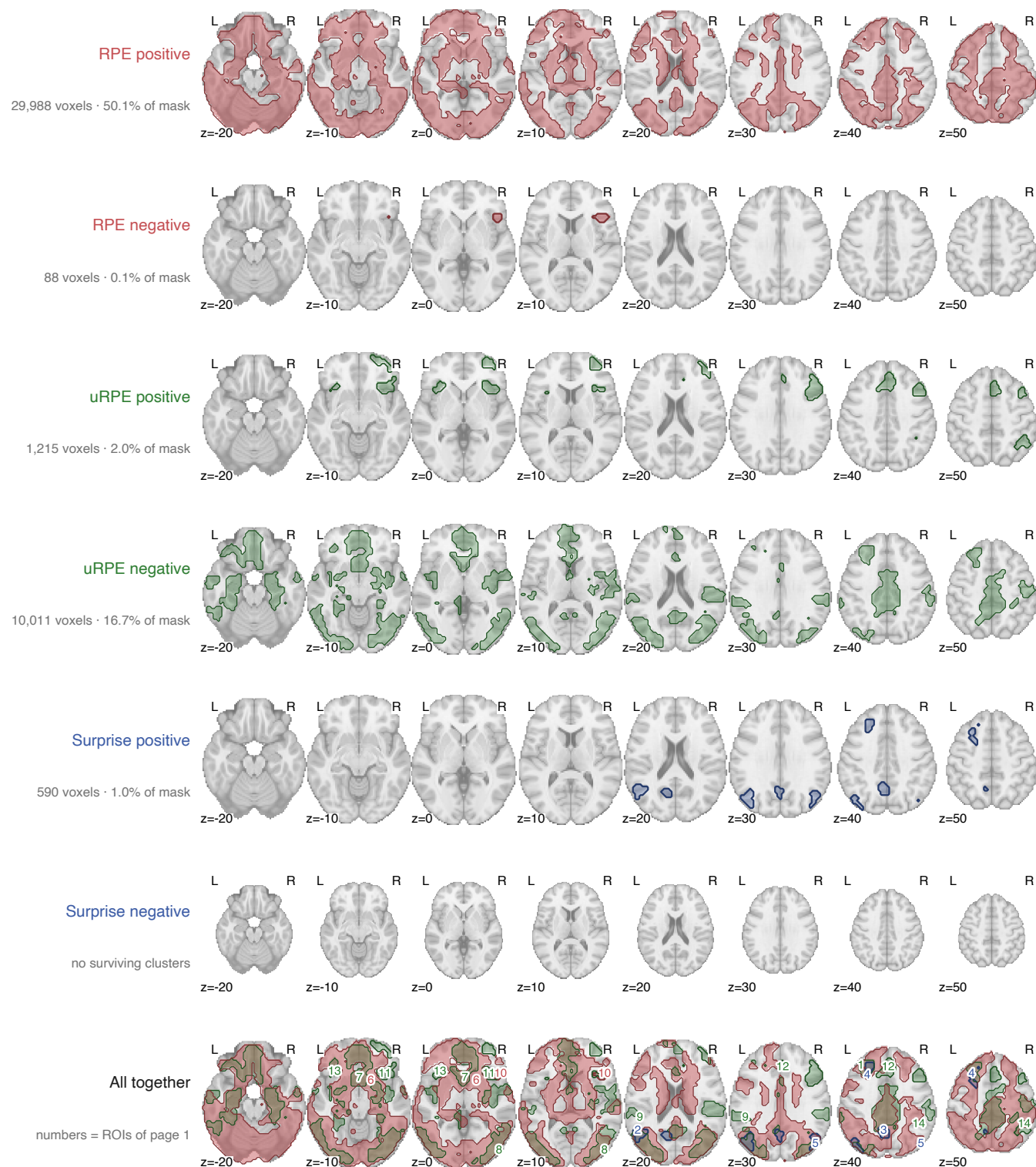
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b Axial sections · $z = -20$ to $+50$ mm

one row per map, then all of them together



Same axial cuts in every row, so extent can be compared straight down a column.

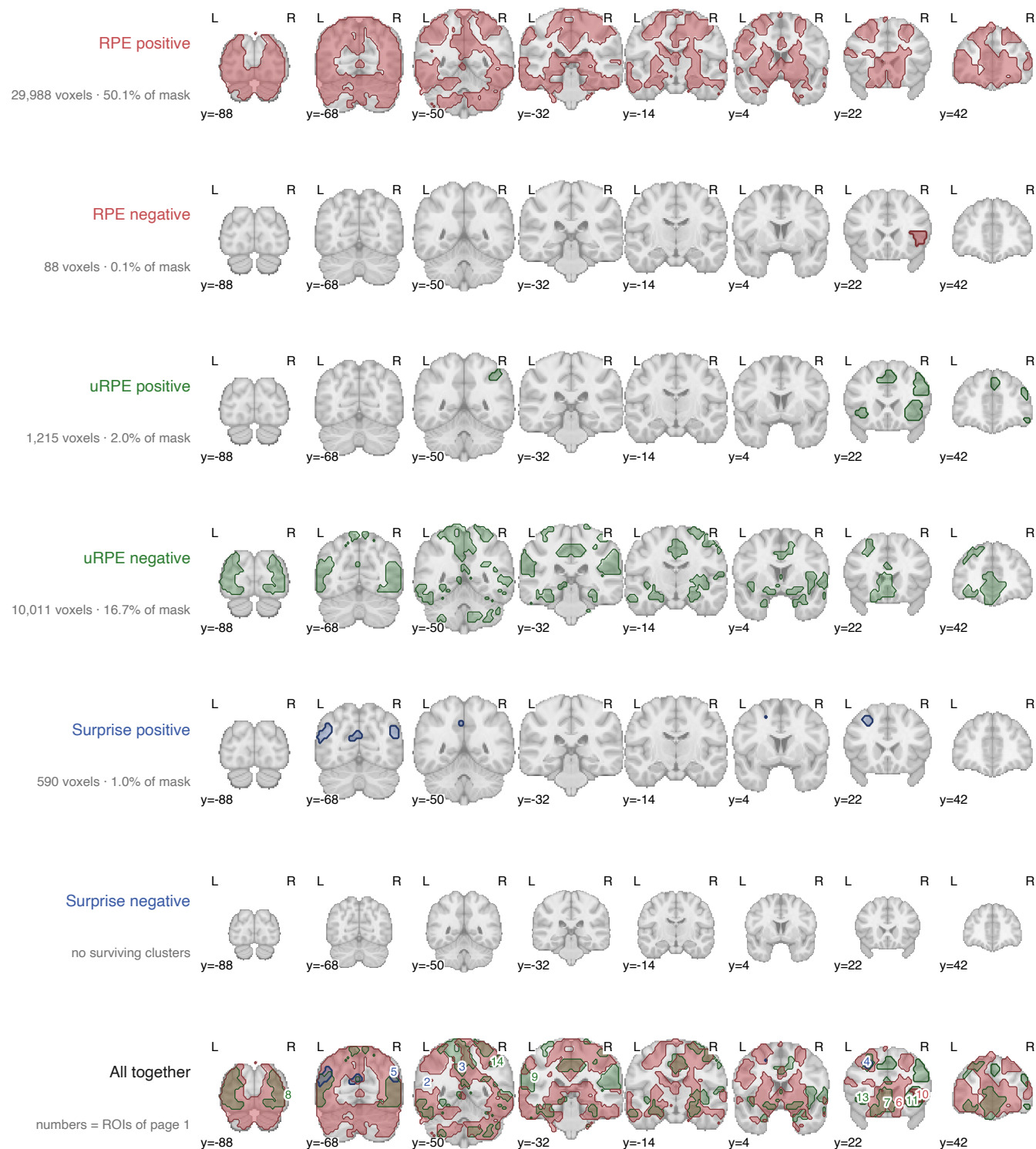
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c Coronal sections · $y = -88$ to $+42$ mm

one row per map, then all of them together



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d Which contrasts claim each piece of tissue

every combination, then the six biggest mapped

